



Four Words on the Job Card: Development and Dyadic Validation of a Two-Form Instrument for Fault-Account Correspondence in Vehicle Service Intake

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Abstract. A customer who books a vehicle in for a fault gives a spoken account of it at the service desk; a service adviser writes that account onto a job card; a technician reads the card. No instrument currently measures how much of the first survives into the third. We report the development and validation of the Intake Account Correspondence Inventory (IACI), a two-form instrument completed independently by the customer (IACI-C) and the adviser (IACI-A) immediately after intake. Forty-one candidate items were generated from 62 recorded intakes, 14 adviser interviews, and an eight-member content-validity panel. Exploratory factor analysis on a calibration half of 604 customer-adviser dyads across nine service centres retained 19 items on three factors — symptom rendering, circumstance retention, and priority framing — accounting for 58.4% of variance. Confirmatory analysis on the holdout half supported the three-factor structure, and internal consistency was adequate on both forms. Metric invariance across forms held; scalar invariance did not, and partial scalar invariance was obtained by freeing two priority-framing intercepts. Dyadic agreement was modest overall and lowest on the subscale the two parties were most confident about. We suggest the instrument is usable in its present form, with the caveat that the two forms are not interchangeable.

Introduction

A vehicle service intake is a short conversation with a long tail. The customer describes a fault — a noise, a smell, a light that comes on sometimes — and the service adviser converts that description into a job card, which is the only version of the account the technician will ever see. The conversion is unremarked and almost entirely unmeasured. Networks that audit intake at all audit its duration.

The service-quality literature has instrumented the encounter itself in considerable depth [1], and the limits of that instrumentation have been rehearsed at length [2]. What those instruments measure is the customer's perception of the service, not the fate of the content the customer supplied. The adjacent problem is better studied in clinical settings, where agreement between

what a patient reports and what a clinician records has been examined directly and found to be low to moderate [3], and where the content a patient never voices at all has been shown to go unrecovered [4]. Maintenance records exhibit the same pattern from the other end: free-text fault descriptions degrade in ways that visibly damage downstream analysis [5], and the tension between structured fields and technician prose is a standing design problem in work-order systems [6].

The job card sits between these literatures. It is a coordinating artefact carried between parties who do not share a vocabulary, in Star and Griesemer's sense [7] — although the term has been applied loosely enough that its author found it necessary to reclaim it [8] — and the labour of keeping it aligned with the actual work

is the articulation work Strauss described [9]. Prior work at this institution has coded the survival of hedges and open questions between an originating account and its minuted form, finding both fall to a median of zero [10].

We make three contributions, each of which we regard as provisional. First, we propose a two-form instrument for fault-account correspondence and report its structural validity on an independent holdout sample. Second, we test measurement invariance across the customer and adviser forms and report where it fails. Third, we relate instrument scores to an independent coding of the job cards themselves, and observe that the relationship is weaker than the instrument's psychometric properties might lead a reader to expect.

Background

Measure development in organisational settings follows a well-established sequence of item generation, structural analysis, and validity assessment [11], with reporting conventions for measurement-property studies codified by the COSMIN panel [12]. We follow both. Internal consistency is reported as ω rather than α , on the grounds that α 's assumptions are rarely satisfied in practice [13], though we report α [14] alongside it for comparability.

Comparing scores across two forms requires evidence that the forms measure the same construct on the same scale. We follow the configural-metric-scalar sequence and the accompanying recommendations for organisational data [15], and report against the conventions surveyed by Putnick and Bornstein [16]. Because our observations are dyadic, we treat the customer-adviser pair as the unit of analysis and report intraclass correlations in the form appropriate to a two-rater, random-effects design [17], [18]; the actor-partner framework [19] informs our treatment of cross-form effects but is not fitted here.

On the technical side, the diagnostic pathway from a reported symptom to a confirmed fault is documented in the professional literature [20]. Our concern is upstream of it.

Method

Item generation

Sixty-two service intakes were audio-recorded across nine centres in a single metropolitan

network, with consent obtained from both parties. Fourteen service advisers were interviewed about what they retain, compress, and drop when completing a card. From these sources 41 candidate items were drafted, each phrased in parallel for the two forms (customer: "I explained when the noise happens"; adviser: "the customer explained when the noise happens"). An eight-member panel rated each item for relevance; 34 items reached an item-level content validity index of 0.86 or above and were retained for field testing.

Sample and procedure

Six hundred and four customer-adviser dyads were recruited at the point of intake over eleven weeks. Both parties completed their form on a tablet immediately after the conversation and before the card was finalised, without sight of the other's responses. Dyads were split at random into calibration ($n = 302$) and holdout ($n = 302$) halves before any analysis was run.

Scoring

The dyad composite is the loading-weighted mean of the two forms' subscale scores:

$$C_d = \sum_{k=1}^3 w_k \cdot \frac{x_{dk}^C + x_{dk}^A}{2}, \quad \sum_{k=1}^3 w_k = 1$$

with $w = (0.42, 0.35, 0.23)$ estimated on the calibration half and held fixed thereafter.

Job-card coding

Independently of the instrument, each dyad's job card was coded against a transcript of its intake. Two raters segmented the customer's account into distinct elements and assigned each element one of four fates on the card: verbatim, paraphrased, reduced to a menu code, or absent. Agreement across 4,472 coded elements was $\kappa = 0.79$. The coding scheme was fixed before the first card was read.

Results

Structure and reliability

Exploratory analysis on the calibration half returned three factors accounting for 58.4% of variance; 15 items were dropped for low or cross-loading, leaving 19. The factors were labelled symptom rendering (8 items), circumstance retention (7 items), and priority framing (4 items). Confirmatory analysis on the holdout half supported this structure: $\chi^2(149) = 271.6$, CFI = 0.961, TLI = 0.953, RMSEA = 0.047 (90% CI 0.039–0.055), SRMR = 0.043. Internal consis-

tency was $\omega = 0.88$ on the customer form and $\omega = 0.86$ on the adviser form ($\alpha = 0.87$ and 0.85 respectively).

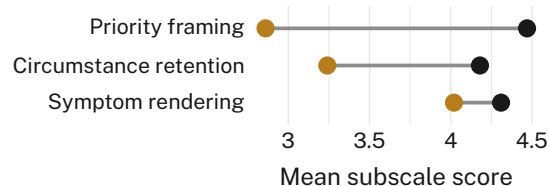


Figure 1: The two forms diverge most where the customer is most confident: mean priority-framing score is 4.47 on the customer form and 2.86 on the adviser form, a gap of 1.61 scale points against 0.29 for symptom rendering.

Invariance across forms

Configural invariance held. Constraining loadings to equality across forms degraded fit negligibly ($\Delta\text{CFI} = 0.004$), supporting metric invariance. Constraining intercepts did not ($\Delta\text{CFI} = 0.017$), exceeding the conventional threshold. Freeing the intercepts of two priority-framing items restored acceptable fit ($\Delta\text{CFI} = 0.006$), giving partial scalar invariance. Latent means may therefore be compared across forms on two of the three subscales, and should not be on the third.

Model	CFI	RMSEA	ΔCFI
Configural	0.961	0.047	—
Metric	0.957	0.049	0.004
Scalar	0.940	0.058	0.017
Partial scalar	0.955	0.050	0.006

Table 1: Scalar invariance is rejected at the conventional 0.010 threshold; partial scalar invariance is obtained by freeing two priority-framing intercepts.

Dyadic agreement

Agreement between the two forms was modest: $\text{ICC}(2,1) = 0.31$ (95% CI 0.24–0.38). Disaggregated by subscale it was 0.52 for symptom rendering, 0.29 for circumstance retention, and 0.12 for priority framing. The ordering is stable across all nine centres.

What reached the card

The independent coding gives a second view of the same intakes. Customers supplied a mean of 7.4 distinct account elements. Of the 4,472 elements coded, 17% appeared on the card verbatim, 25% in paraphrase, 22% as a menu code with no free text, and 36% not at all (Figure 2). Elements the customer had volunteered without

being asked were absent at roughly twice the rate of elements given in answer to a question.

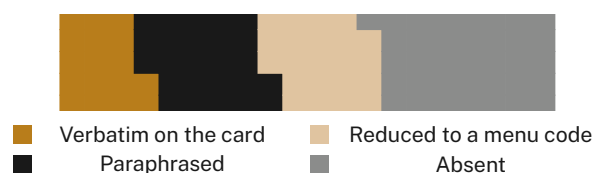


Figure 2: Just over a third of what the customer says reaches no field on the card in any form, and a further 22% survives only as a menu code.

The composite correlated with the proportion of elements surviving in any form at $r = 0.24$ ($p < 0.05$). The instrument and the card are therefore measuring related but largely separate things.

Ablation

Removing the four-item priority-framing subscale — the subscale on which scalar invariance failed and agreement was lowest — changes the composite by 0.4% and model fit by $\Delta\text{CFI} = 0.002$, neither significant (Figure 3). We retain the subscale for completeness.

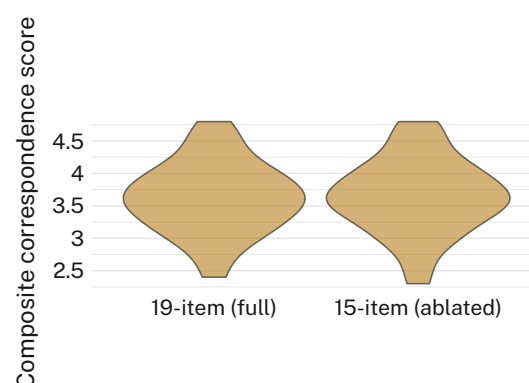


Figure 3: Composite score distributions under the 19-item and 15-item scorings are statistically indistinguishable (mean difference 0.4%, n.s.).

Discussion

The instrument behaves. Its structure replicates on a holdout sample, its reliability is adequate on both forms, and its factors correspond to distinctions advisers themselves draw. What it does not do is agree with itself across the two parties, and the disagreement is systematic rather than noisy: it is largest on priority framing, where the customer's confidence that the point was made and the adviser's confidence that it was received are, on these data, close to unrelated.

This has a practical consequence for how the instrument is deployed. The network hosting this study has adopted the adviser form alone for monthly reporting, on the reasonable ground that it is the cheaper of the two to collect. The adviser form is also the one whose scores agree least with the customer's, and the resulting figure has since been listed by the Indicator Commons as a candidate intake measure. Nothing in our results supports reading a single-form score as a correspondence score, and we note the practice here without taking a position on it. Consistent with the finding of Adegoke and Hanke that a record built for one purpose acquires a second use it was not designed to carry [21], the instrument appears to be following its own score into an application we did not anticipate.

Limitations

The study draws on a single metropolitan network, though the consistency of the subscale ordering across all nine of its centres suggests the pattern is not centre-specific. Both forms are self-report, and neither is validated against a technician-side measure; that the two nonetheless disagree so regularly may indicate the disagreement is conservative rather than inflated. Job-card coding was performed by two raters rather than a larger panel, although agreement was high enough that additional raters would have had little to disagree with.

Conclusion

We present a two-form instrument for fault-account correspondence at vehicle service intake, with acceptable structural validity, adequate reliability, and partial scalar invariance across forms. Dyadic agreement is modest and lowest where both parties are most confident, and instrument scores track only weakly onto what independent coding finds on the job card itself. We suggest the two forms be reported as a pair. Future work should establish whether the pattern holds where the intake is taken by telephone rather than at a desk.

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